Predicting STEM Undergraduates' Intentions to Pursue Scientific Research Careers

Motivation

For decades, policymakers, educators, and national commissions have called for expanding the number of sciences, technology, engineering, and mathematics (STEM) degrees produced in the U.S., particularly for graduate degrees (National Science Board, 2020; President's Council of Advisors on Science and Technology, 2012). Understanding the factors that influence students' intentions to pursue scientific research careers is important to cultivate the next generation of scientists and researchers. A growing body of research has examined early, scalable interventions, such as faculty mentorship and undergraduate research programs, that can strengthen students' interest, skills, and preparation to pursue scientific research careers (Bruthers et al., 2021; Eagan et al., 2024; Estrada et al., 2018). While many of these studies have relied primarily on linear regression approaches, fewer have explored potential non-linear relationships using nonparametric methods or compared traditional statistical techniques with modern machine learning algorithms. Leveraging the Higher Education Research Institutions (HERI)'s 2019 College Senior Survey (CSS), this study aims to address the following research questions:

RQ1: How do scientific career aspirations, undergraduate research time, and time allocation patterns vary across different STEM majors?

RQ2: What relationships exist between predictors and scientific research career aspirations? Do students' undergraduate research experiences correlate with science-related psychological traits?

RQ3: How accurately can machine learning algorithms (logistic regression, decision trees, random forests, K-nearest neighbors, and XGBoost) predict students' likelihood of pursuing science research careers, and which features are most important across different models?

Methods

Data Source and Sample

The data used in this study from HERI's 2019 CSS, which was administered in the last quarter/semester when undergraduate seniors going to graduate from universities, collects students' college experiences, psychosocial outcomes, and post-college plans information. The analytic sample includes students who graduate with a STEM bachelor's degree. The final analytic sample consisted of 3,346 STEM undergraduate students. Among these students, around 37% from Health Professions, 15% from Biological & Life Sciences, 23% from Math & Computer Science, 7 % from Engineering, and 17 % from Physical Science. The major classification code is shown in Appendix A.

Missing data analysis revealed that the overall missingness rate was less than 1.6% across all variables. Little's Missing Completely at Random (MCAR) test indicated that the missingness pattern was not systematically related to observed values ($p = 0.567$), suggesting that listwise deletion was appropriate. For continuous psychological construct variables (e.g., science self-efficacy, science identity, academic self-concept, and habits of mind), I identified and removed 22 outliers defined as cases with standardized scores exceeding $|z| > 3.0$. Ordinal variables measured on Likert scales (3-5 points) were not subject to outlier removal or Winsorization.

Variables

The outcome variable I want to predict in this study is students' scientific career aspirations. In the survey, students reported their intentions to pursue a science-related research career on a 5-point ordinal scale (1 = Definitely no, 2 = Probably no, 3 = Uncertain, 4 = Probably yes, 5 = Definitely yes). For machine learning classification analyses, this variable was dichotomized into a binary indicator where responses "Probably yes" and "Definitely yes" were coded as 1 (intends to pursue a scientific research career), and all other responses were coded as 0.

The key variable of interest is students undergraduate research experience, which is measured by duration of participation in a professor’s research project during college on a six-point scale (1 = 0 months, 2 = 1-3 months, 3 = 4-6 months, 4 = 7-12 months, 5 = 13-24 months, 6 = 25+ months). Other covariates include eight-point scaled students’ major GPA (from 1 = D to 8 = A/A+), continuous psychological constructs (i.e., science self-efficacy, science identity, academic self-concept, and habits of mind), four-point scaled goal orientation variables (i.e., aiming to make theoretical contributions to science, helping others who are in difficulty, and becoming an authority in the field), as well as eight-point scaled college time allocation variable, such as time spend on studying/homework, socializing with friends in person, using social media, etc. (from 1 = None to 8 = more than 20 hours). The full list of variables is presented in Appendix B.

Analysis

This study employed a multi-method approach to address the research questions, including exploratory descriptive visualization, linear regression, kernel regression, and machine learning classification. To begin with, I created several box plots with overlaid jittered raw data points to visualize the distribution of scientific research career aspirations and the time work on professors’ research projects across different STEM majors. Additionally, I examined patterns in time allocation across different activities by major discipline.

To examine relationships between predictors and scientific research career aspirations, I fitted an ordinary least squares (OLS) linear regression model with scientific research career aspirations as the continuous outcome and all predictor variables as independent variables.

Model diagnostics, including residual plots, Q-Q plots, and tests for heteroscedasticity and normality, were conducted to assess model assumptions. Moreover, to explore potential non-linear relationships, I implemented Nadaraya-Watson kernel regression using a Gaussian kernel with Scott's rule bandwidth selection. This nonparametric approach was implemented in C++ using the Rcpp function in R for computational efficiency. Bootstrap resampling (B = 200 iterations) was employed to construct 95% confidence intervals around the kernel regression estimates. I examined kernel regression relationships between undergraduate research experience with STEM students' science identity, science self-efficacy, academic self-concept, and habits of mind.

To predict students' binary scientific research career intentions, I implemented five machine learning algorithms: logistic regression, decision trees (CART), random forests, K-nearest neighbors (KNN), and gradient boosting (XGBoost). All models were trained using 10-fold cross-validation to ensure robust performance estimates and prevent overfitting (James et al., 2013). Model performance was evaluated using three measures calculated across 10 folds: (1) Receiver Operating Characteristic Area Under the Curve (ROC-AUC), which measures the model's ability to discriminate between students who will and will not pursue scientific research careers; (2) Sensitivity (true positive rate), indicating the proportion of students intending to pursue scientific research careers who are correctly identified; and (3) Specificity (true negative rate), representing the proportion of students not intending to pursue scientific research careers who are correctly classified.

Variable importance was assessed for models (i.e., logistic regression, decision trees, random forests, and XGBoost) using model-specific importance measures. For logistic regression, importance was derived from standardized coefficients representing direct linear effects. Decision trees calculated importance based on reduction in node impurity (Gini index) when variables were used for splits. Variables that create the “purest” splits (best separate the classes) are most important. Random forests averaged impurity reduction across all trees in the ensemble, while XGBoost measured gain, the average improvement in accuracy when variables were used for splitting across sequentially trained trees. These different approaches capture distinct aspects of variable influence: logistic regression identifies independent main effects, single decision trees detect non-linear patterns, and ensemble methods (random forests and XGBoost) incorporate interaction effects, potentially distributing importance among correlated or interacting predictors. All importance scores were scaled to a 0-100 range, where 100 represents the most important predictor. I analyzed both individual model importance rankings and combined importance scores across models to determine the relative importance of undergraduate research experiences in predicting students’ scientific research career aspirations.

Findings

Descriptive Analysis Across STEM Disciplines

Figure 1(a) presents the distribution of scientific research career aspirations and durations of undergraduate research experience, and Figure 1(b) shows the box plots of the hours per week students spent on activities. Students who majored in Health Professions and Engineering appear to have relatively higher means in pursuing scientific research careers, and they also seem to report relatively higher average time of undergraduate research experience with faculty compared to other majors. In terms of time allocation patterns, almost all majors reported spending six to ten hours per week on studying and homework. However, students’ time allocation on studying with other students and using social media seems to show notable variations. Math and computer science, and physical science students tend to devote more time to studying with peers, while students who majored in Heath professions and physical science appear to allocate higher average time to social media.

Linear and Kernel Regression Results

Table 1 presents the estimated results of the linear regression model, which explained 36.1% of the variance in scientific research career aspirations (adjusted $R^2 = 0.361$, $F(18, 3327) = 106$, $p < 0.001$). The standardized coefficient shows that science identity demonstrated the largest effect size compared to other predictors ($\beta = 0.37$, $p < 0.001$), indicating that a one standard deviation increase in science identity was associated with a 0.37 standard deviation increase in scientific research career aspirations. Students who have a strong commitment to make theoretical contributions to science (Goal19) also have significantly higher aspirations for scientific research careers ($\beta = 0.28$, $p < 0.001$). Students who reported longer during of doing undergraduate research experience with faculty demonstrated a significant positive association with aspirations towards a scientific research career ($\beta = 0.10$, $p < 0.001$), after controlling for other variables. Interestingly, students who committed to help others in difficulty showed a significantly negative association with scientific research career pursuit ($\beta = -0.07$, $p < 0.001$). Model diagnostic plots, showed in Figure 2, seem to reveal a generally acceptable model fit, with residuals appearing approximately normally distributed around zero and no severe violations of homoscedasticity. However, the Q-Q plot showed some deviation in the tails, which may indicate slight departures from normality that are common with ordinal outcome variables.

Figure 3 presents kernel density estimates with bootstrap confidence intervals for the four continuous latent psychological constructs. Science identity, science self-efficacy, and habits of mind exhibited bimodal or multimodal distributions, suggesting the presence of different student subgroups with different levels of psychological traits. Kernel regression analyses (Figure 4) revealed that undergraduate research experience exhibited positive monotonic relationships with all four psychological constructs. Students who had longer undergraduate research experience with faculty showed the strongest gains in science identity, with steeper increases during initial and moderate research durations (0-12 months) before plateauing at extended involvement (13+ months). Students' science self-efficacy, academic self-concept, and habits of mind showed more modest, approximately linear increases with research duration. These nonparametric results indicate that while students who engage in more research consistently demonstrate stronger psychological outcomes, the magnitude and shape of these developmental trajectories differ across constructs.

Machine Learning Classification and Variable Importance Analysis

Figure 5 presents the comparative performance of five machine learning algorithms in predicting binary scientific research career aspirations. Across 10-fold cross-validation, the XGBoost and logistic regression models demonstrated the highest overall performance, with mean ROC-AUC values is 0.810 and 0.808, respectively. Moreover, KNN achieved the highest sensitivity (mean = 0.804), correctly identifying 80% of students who intend to pursue research careers, but at the cost of lower specificity (mean = 0.591). Conversely, XGBoost yields the highest specificity (mean = 0.704), meaning it was better at avoiding incorrectly predicting that students would pursue research careers when they actually would not.

Figure 6 and Table 2 present the variable importance analysis across different algorithms. As we can see in Figure 6, science identity yields the highest maximum importance score (100) in logistic regression, decision trees, random forests, and XGBoost. This convergence highlights science identity as the most powerful predictor of STEM undergraduates' scientific research career intentions, which corroborates the linear regression findings. Students' goal to make theoretical contributions to science (Goal19) ranked as the second most important predictor, showing high importance in logistic regression (85.2) and decision trees (96.6), but lower importance in random forest (66.1) and XGBoost (43.4). This pattern suggests that the variable operates primarily as a direct predictor, while ensemble methods likely distribute its predictive power across interactions with other variables or correlated predictors. Research experience ranked third in average importance across models (mean = 32.7), with consistent contributions in logistic regression (45.1), decision trees (47.8), and random forests (26.7).

Notably, students' goal to help others in difficulty showed large importance differences across the four models, with high importance in logistic regression (32.2) but minimal in tree-based methods (< 3.0). This discrepancy may stem from how algorithms measure importance. Logistic regression may identify it as a significant predictor that explains variance after controlling for other variables, while tree-based methods assign low importance because this goal may not create effective splits for distinguishing between students who will and will not pursue research careers. In other words, this may indicate that its predictive power may depend on other variables, and the primary predictive signal is already captured by science identity and other goal orientations.

Conclusion and Discussion

This study examined factors that predict STEM undergraduates' aspirations to pursue scientific research careers, focusing particularly on undergraduate research experiences and psychological constructs. Results showed that three factors were positively associated with intentions to pursue a scientific research career, including science identity, students' aspirations to make theoretical contributions to science, and the amount of time spent conducting undergraduate research with faculty. While traditional statistical methods remain valuable and can easily be interpreted, the machine learning analyses demonstrate that ensemble algorithms like XGBoost can achieve relatively higher predictive performance by capturing complex interactions among predictors.

When interpreting these findings, several limitations should be acknowledged. First, the cross-sectional nature of the CSS data precludes causal inference about the relationship between undergraduate research experiences and career aspirations. Students with stronger initial scientific research career interests may self-select into research opportunities. Second, the study relies on self-reported data, students may have over- or under-reported their research involvement or inflated their science identity and career aspirations. Third, the binary classification of career aspirations, while necessary for machine learning approaches, may oversimplify the career decision-making process. Finally, this study only limited to students who graduated with a STEM bachelor's degree, which may not capture students who left STEM fields during their undergraduate studies.

Moving forward, future study could use quasi-experimental designs, such as propensity score matching or instrumental variable approaches, to better assess the causal effects of undergraduate research participation on scientific research career development. Longitudinal studies tracking students from college entry through early career stages would provide insights into how research experiences shape career trajectories over time. Additionally, future studies might expand beyond student-level variables to include institutional-level factors, such as institution type (research universities vs. community colleges), selectivity level, whether the institution serves as a Historically Black Colleges or Universities (HBCU) or Hispanic-Serving Institution (HSI), proportion of STEM bachelor's degrees awarded, and student-faculty ratio. Incorporating these institutional characteristics would enable multilevel modeling to assess how institutional contexts moderate the relationship between undergraduate research participation and scientific research career aspirations. Furthermore, future research could examine whether the importance of key predictors varies across different students' populations and institutional contexts. Finally, future studies might also examine the heterogeneous effects of undergraduate research experiences on students' career aspirations across gender, racial/ethnic, and socioeconomic identities. Such work can advance our understanding of how students convert early academic research opportunities into sustained scientific research careers.

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Table 1. Linear Regression Results Predicting Scientific Research Career Aspirations

Variable	B	β	SE	95% CI [LCI, UCI]		p-value
(Intercept)	2.350	NA	0.166	2.020	2.680	<.001***
Science self-efficacy (z)	0.035	0.021	0.029	-0.022	0.092	.229
Science identity (z)	0.615	0.366	0.032	0.554	0.677	<.001***
Academic self-concept (z)	-0.111	-0.085	0.022	-0.154	-0.068	<.001***
Habits of mind (z)	-0.016	-0.011	0.022	-0.059	0.028	.482
Major GPA	-0.020	-0.024	0.014	-0.047	0.007	.137
Make theoretical contribution to science (GOAL19)	0.370	0.277	0.023	0.324	0.416	<.001***
Helping others who are in difficulty (GOAL11)	-0.117	-0.067	0.025	-0.167	-0.067	<.001***
Goal: Becoming an authority in my field (GOAL04)	-0.016	-0.010	0.024	-0.063	0.031	.502
Undergraduate Research (RESEARCH)	0.072	0.099	0.011	0.050	0.094	<.001***
Studying/homework (HPW17)	0.037	0.041	0.014	0.009	0.065	.009**
Studying with other students (HPW29)	-0.004	-0.004	0.014	-0.030	0.023	.796
Socializing with friends in person (HPW28)	-0.049	-0.057	0.013	-0.075	-0.022	<.001***
Using social media (HPW09)	-0.010	-0.012	0.013	-0.035	0.016	.464
Participating in student clubs/groups (HPW04)	-0.019	-0.024	0.012	-0.043	0.004	.112
Watching TV/online video content (HPW24)	0.010	0.011	0.013	-0.016	0.035	.468
Exercising/sports (HPW06)	0.012	0.015	0.012	-0.011	0.036	.305
Working (for pay) on campus (HPW27)	0.005	0.008	0.009	-0.013	0.022	.595
Career planning (job searches, networking)(HPW02)	0.006	0.006	0.015	-0.024	0.037	.684

Note: N = 3,346. B = unstandardized coefficient; β = standardized coefficient; SE = standard error; CI = confidence interval; LCI = lower confidence interval; UCI = upper confidence interval. All psychological constructs are z-scored. ***p < .001. **p < .01. *p < .05.

Table 2. Variable Importance Scores Across Different Algorithms

Variable	**Logistic Regression**	**Decision Tree**	**Random Forest**	**XGBoost**	**Average**
Science Identity	100.00	100.00	100.00	100.00	100.00
Make theoretical contribution to science (GOAL19)	85.20	96.60	66.10	43.40	72.80
Undergraduate Research (RESEARCH)	45.10	47.80	26.70	11.20	32.70
Academic self-concept	30.20	19.20	12.20	10.60	18.10
Science self-efficacy	8.63	34.30	20.40	5.04	17.10
Habits of mind	1.08	22.00	10.70	6.75	10.10
Major GPA	12.20	16.90	5.95	4.03	9.79
Helping others who are in difficulty (GOAL11)	32.20	0.55	0.00	3.64	9.09
Socializing with friends in person (HPW28)	19.50	5.51	6.18	3.50	8.68
Working (for pay) on campus (HPW27)	9.28	4.95	8.09	0.78	5.78

Figure 1(a) Distribution of Scientific Research Career Aspirations and Undergraduate Research Experience Duration by Major

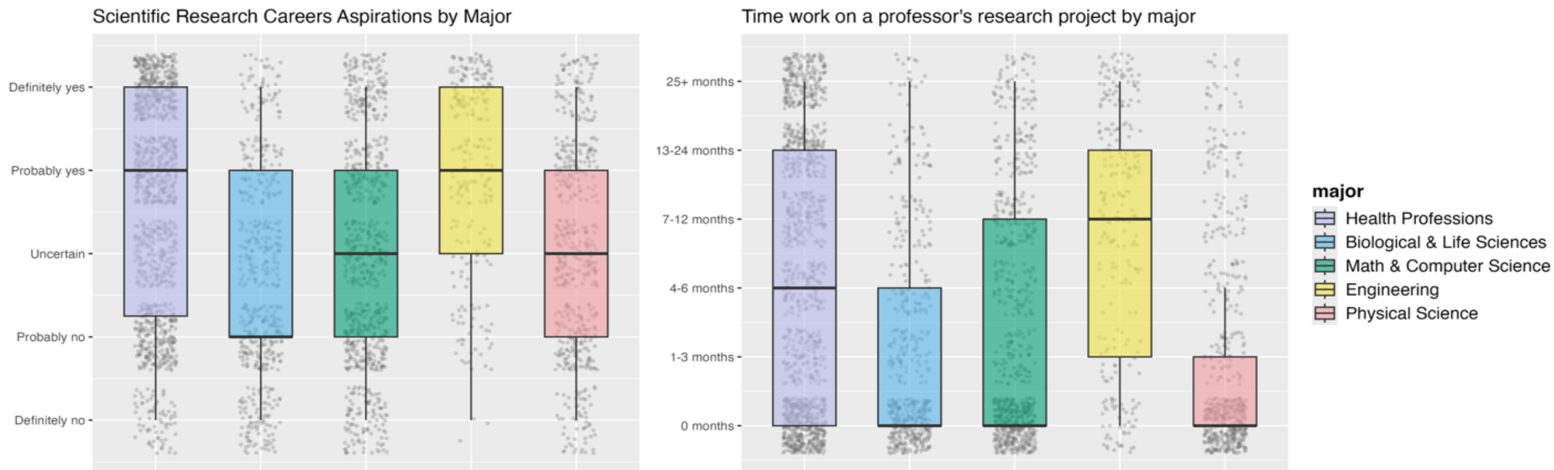


Figure 1(b) Weekly Time Allocation Patterns Across Major

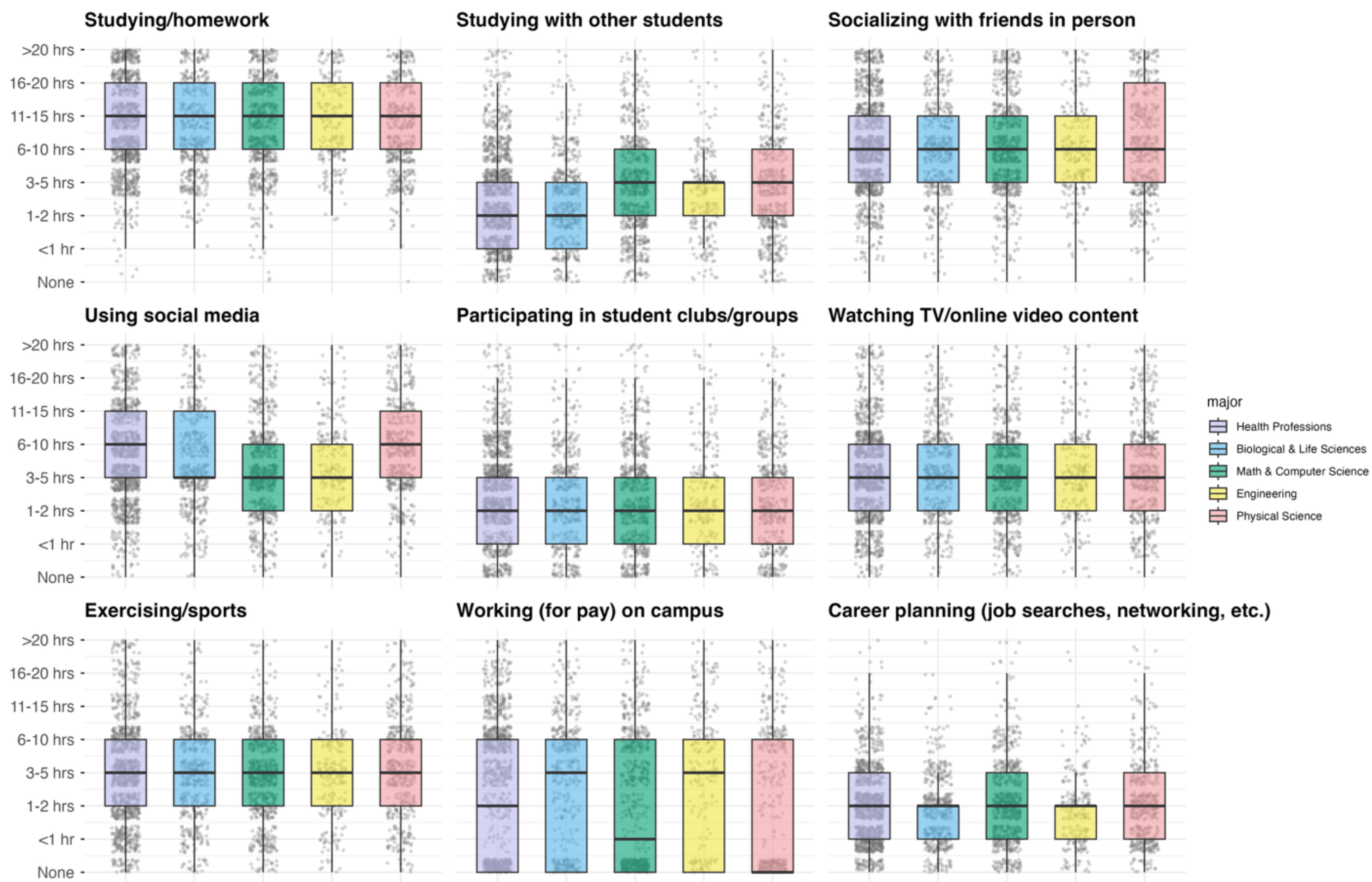


Figure 2. Diagnostic Plots for Linear Regression Model Assumptions

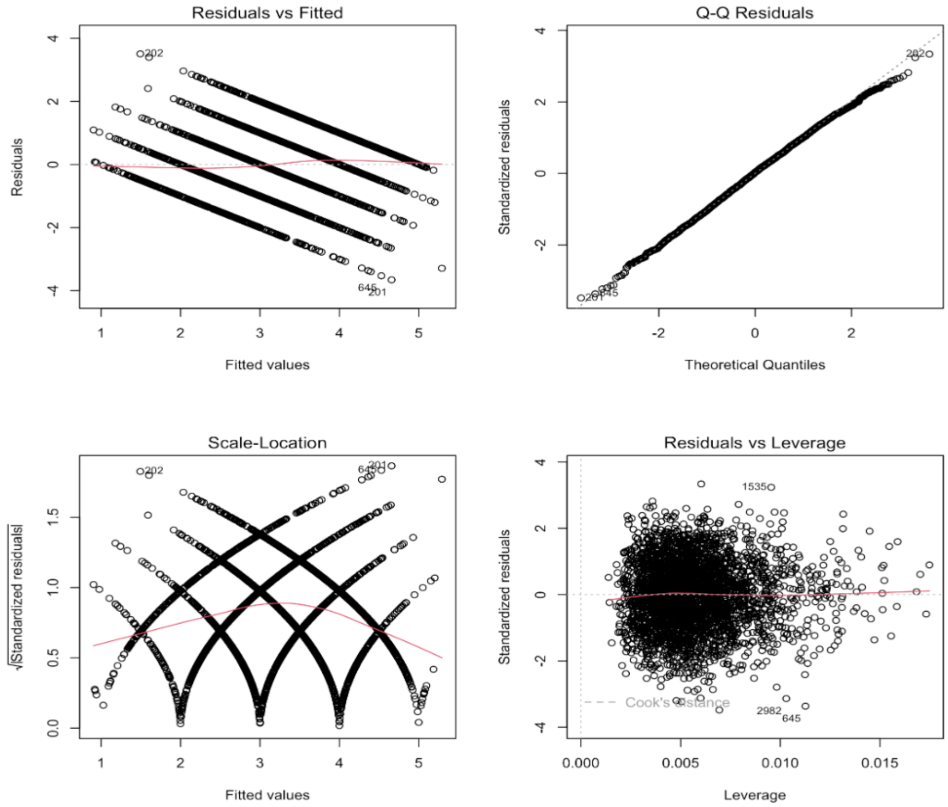


Figure 3. Kernel Density Estimates of Psychological Constructs with Bootstrap Confidence Intervals

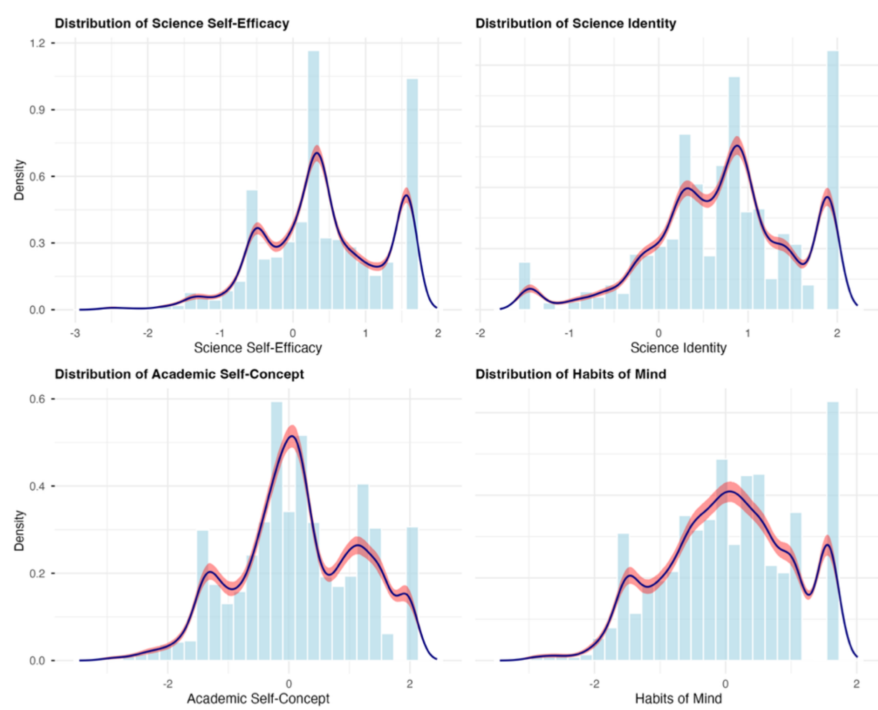


Figure 4. Relationship Between Undergraduate Research Experience Duration and Psychological Constructs (Kernel Regression Analysis)

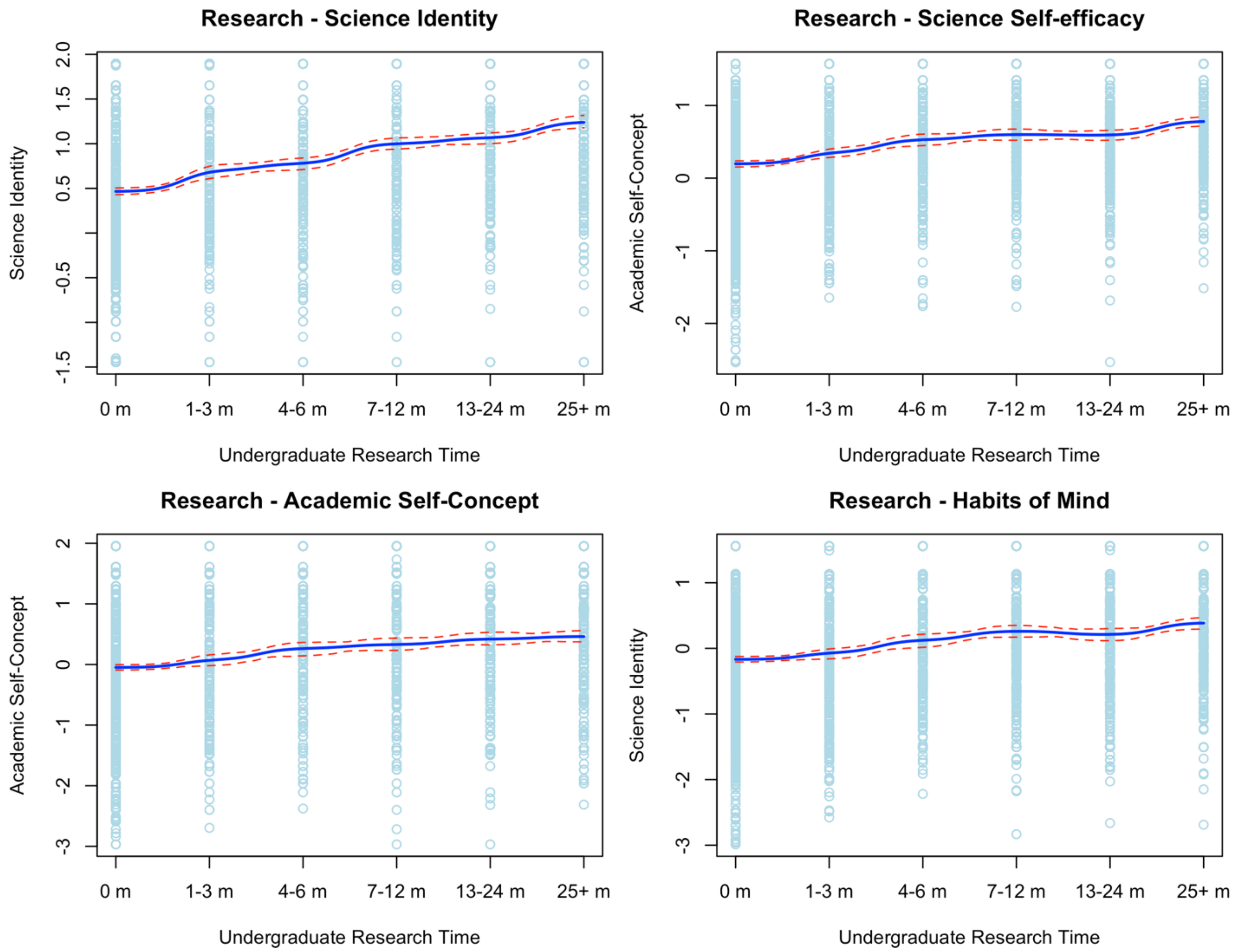


Figure 5. Performance of ML Algorithms in Predicting Scientific Research Career Aspirations

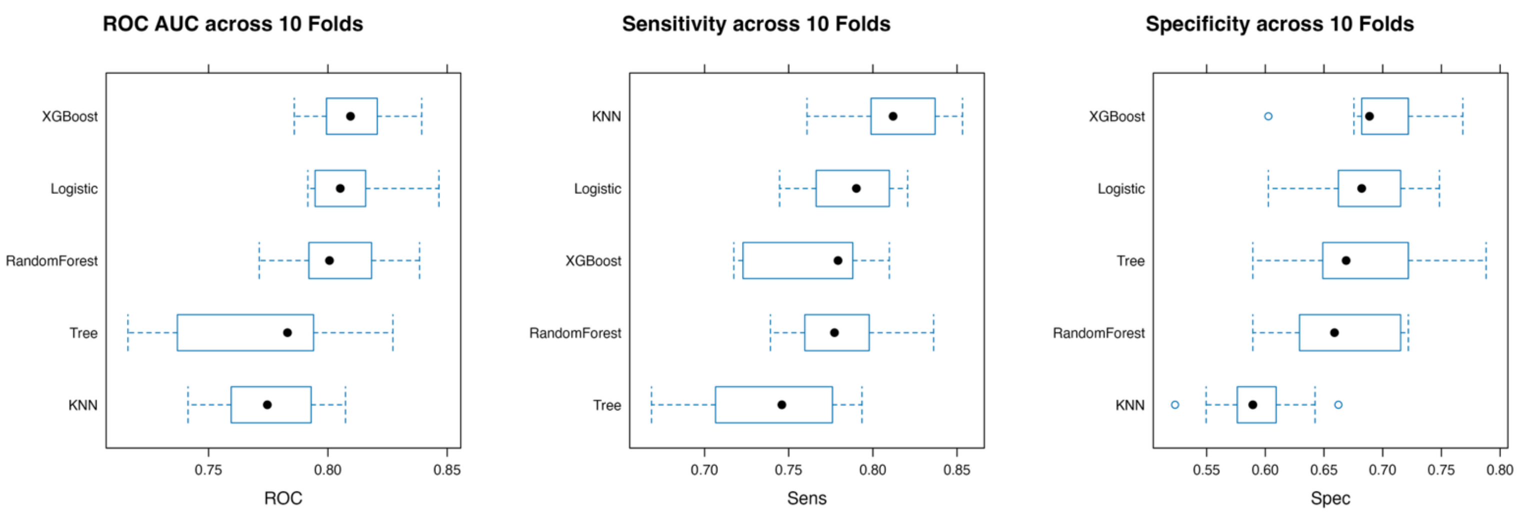
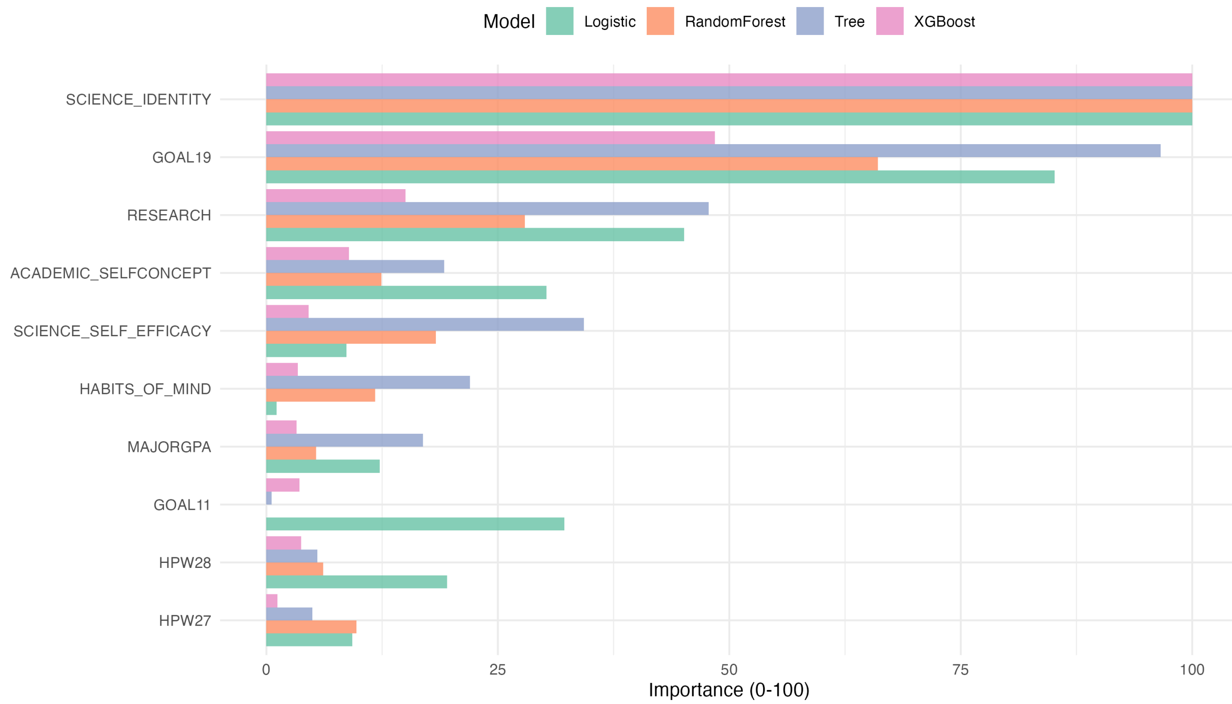


Figure 6. Variable Importance Rankings (Top 10) Across Different Algorithms



Appendix A. STEM Major Classification in TFS and CSS

Biological & Life Sciences	Agriculture, Agriculture/Natural Resources, Animal Biology (zoology), Biochemistry/Biophysics, Biology (general), Botany, Ecology & Evolutionary Biology, Environmental Science, Marine (life) Science, Marine Biology, Microbiology, Microbiology or Bacteriology, Molecular Cellular & Developmental Biology, Neurobiology/Neuroscience, Other Biological Science, Plant Biology (botany), Zoology
Engineering	Aeronautical or Astronautical Eng, Aerospace/Aeronautical/Astronautical Engineering, Biological/Agricultural Engineering, Biomedical Engineering, Chemical Engineering, Civil Engineering, Computer Engineering, Electrical or Electronic Engineering, Electrical/Electronic Communications Engineering, Engineering Science/Engineering Physics, Environmental/Environmental Health Engineering, Industrial Engineering, Industrial/Manufacturing Engineering, Materials Engineering, Mechanical Engineering, Other Engineering
Math & Computer Science	Computer Science, Data Processing or Computer Programming, Mathematics, Mathematics/Statistics, Other Math and Computer Science, Statistics
Physical Science	Astronomy, Astronomy & Astrophysics, Atmospheric Sciences, Chemistry, Earth & Planetary Sciences, Earth Science, Marine Sciences, Other Physical Science, Physics
Health Professions	Clinical Laboratory Science, Health Care Administration/Studies, Health Technology, Medical Dental Veterinary, Nursing, Other Health Profession, Other Professional, Pharmacy, Therapy (occupational physical speech)

Appendix B. Variable List

Variable	Coding Description
Dependent Variables	
Scientific research career aspirations (SCICAREER)	Will you puruse a science-related research career? 1=Definitely no; 2=Probably no; 3=Uncertain; 4=Probably yes; 5=Definitely yes
Scientific research career aspirations (SCICAREER)	Binary: 1 = Yes (probably/definitely yest); 0 = Otherwise
Predictors	
Undergraduate Research Experience (RESEARCH)	How many months since entering college (including summer) did you work on a professor's research project? 1=0 months; 2=1-3 months; 3=4-6 months; 4=7-12 months; 5=13-24 months; 6=25+ months
Students' major GPA (MAJORGPA)	1=D; 2=C; 3=C+; 4=B-; 5=B; 6=B+; 7=A-; 8=A or A+
Science self-efficacy	A continous construct of students' confidence in their ability to conduct scientific research.
Science identity	A continous contrust describing the extent to which students conceive of themselves as scientists.
Academic self-concept	A continous construct of students' beliefs about their abilities and confidence in academic environments.
Habits of mind	A continuous construct of the behaviors and traits associated with academic success. These behaviors are seen as the foundation for lifelong learning.
Goal: Making a theoretical contribution to science (GOAL19)	1=Not Important; 2=Somewhat Important; 3=Very Important; 4=Essential
Goal: Helping others who are in difficulty (GOAL11)	1=Not Important; 2=Somewhat Important; 3=Very Important; 4=Essential
Goal: Becoming an authority in my field (GOAL04)	1=Not Important; 2=Somewhat Important; 3=Very Important; 4=Essential
Hours per Week: Studying/homework (HPW17)	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 20
Hours per Week: Studying with other students	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 21
Hours per Week: Socializing with friends in person	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 22
Hours per Week: Using social media	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 23
Hours per Week: Participating in student clubs/groups	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 24
Hours per Week: Watching TV/online video content	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 25
Hours per Week: Exercising/sports	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 26
Hours per Week: Working (for pay) on campus	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 27
Hours per Week: Career planning (job searches, networking, etc.)	1=None; 2=Less than 1 hour; 3=1-2; 4=3-5; 5=6-10; 6=11-15; 7=16-20; 8=Over 28